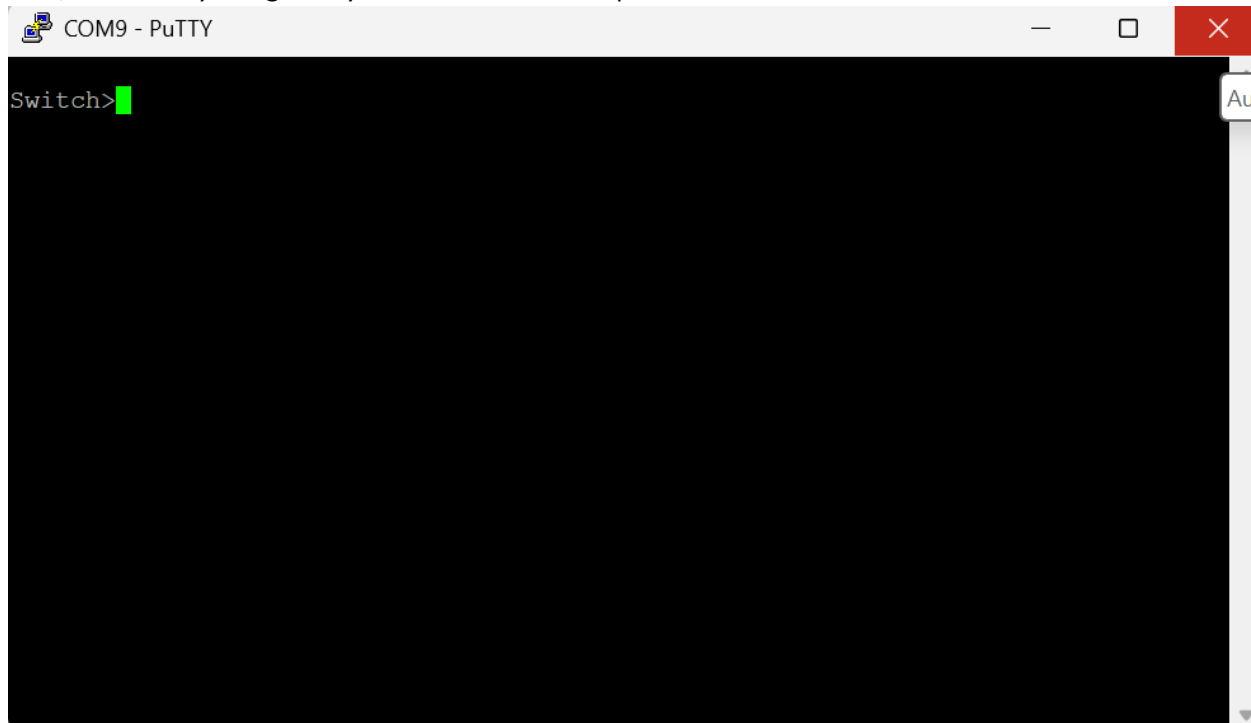


Networks are often segmented poorly, and allow positions to access data they have no need for. This allows an attacker, once compromising a system, to have unrestricted access to many departments and their data. This project seeks to mitigate this threat through quality network segmentation, VLAN routing, and firewall deny rules.

Project objective: Prevent common lateral movement attacks inside of a network enterprise utilizing VLAN segmentation, inter-VLAN routing, and firewall enforced default deny policies. Valid effectiveness through controlled attack simulation.

1. To start, I will be using the L3 switch, enable Layer 3 routing, and to create the VLANs. I will then create the SVI gateways and test their effectiveness.
 - a. First, I Started by using PuTTY and enter the COM9 port. I entered the terminal.



- b. After I entered a series of commands to start creating the VLANS we want to protect. I entered "Enable" to enter EXEC mode, which allows me access configuration modes. Then entered "configure terminal" to start altering device configurations. I then entered "ip routing" to enable VLANS to communicate with each other by allowing the switch to act as an internal router.

```

Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#ip routing
Switch(config)#

```

next, I create the VLANs themselves.

```

Switch(config)#vlan 10
Switch(config-vlan)#name HR
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name FINANCE
Switch(config-vlan)#exit
Switch(config)#vlan 30
Switch(config-vlan)#name IT
Switch(config-vlan)#exit
Switch(config)#vlan 40
Switch(config-vlan)#name SALES
Switch(config-vlan)#exit
Switch(config)#vlan 50
Switch(config-vlan)#name DMZ
Switch(config-vlan)#exit
Switch(config)#

```

Able to confirm them in the switch terminal

VLAN	Name	Status	Ports
1	default	active	Gil/0/8, Gil/0/9, Gil/0/10 Gil/0/11, Gil/0/12, Gil/0/13 Gil/0/14, Tel/0/1, Tel/0/2
10	HR	active	Gil/0/1
20	FINANCE	active	Gil/0/2, Gil/0/3, Gil/0/4
30	IT	active	Gil/0/5, Gil/0/6, Gil/0/7
40	SALES	active	
50	DMZ	active	

- c. Next, I must create the SVI gateways allow communication outside of the VLANs.

```
ged state to downexit
switch(config)#interface vlan 10
switch(config-if)#ip address 10.10.10.1 255.255.255.0
switch(config-if)#no shutdown
switch(config-if)#exit
switch(config)#interface vlan 20
switch(config-if)#
ful 3 21:48:21.764: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan20, cha
ged state to downip address 10.10.20.1 255.255.255.0
      ^
Invalid input detected at '^' marker.

switch(config-if)#ip address 10.10.20.1 255.255.255.0
switch(config-if)#no shutdown
switch(config-if)#exit
switch(config)#interface vlan 30
switch(config-if)#
ful 3 21:49:00.282: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan30, cha
ged state to downip address 10.10.30.1 255.255.255.0
switch(config-if)#exit
switch(config)#
```

Important to segment them properly (10.10.10.1, then 10.10.20.1, etc) to separate them as distinct vlans.

- d. Next I need to configure physical ports for the VLANs. Setting “switchport mode access” gives this port access to traffic, and setting it to “access vlan x” gives only that VLAN’s traffic

```

Switch(config)#interface gigabitEthernet 1/0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface gigabitEthernet 1/0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface gigabitEthernet 1/0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#exit
Switch(config)#interface gigabitEthernet 1/0/4
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 40
Switch(config-if)#exit
Switch(config)#interface gigabitEthernet 1/0/5
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 50
Switch(config-if)#exit
Switch(config)#

```

I also created a trunk port on port 10.

```

Switch(config)#interface gigabitEthernet 1/0/10
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk allowed vlan 10,20,30,40,50
Switch(config-if)#exit
Switch(config)#

```

Using the show command, we can see everything is functional.

```

VLAN Name                               Status    Ports
-----
1      default                             active    Gi1/0/8, Gi1/0/9, Gi1/0/10
                                           Gi1/0/11, Gi1/0/12, Gi1/0/13
                                           Gi1/0/14, Tel/0/1, Tel/0/2
10     HR                                 active    Gi1/0/1
20     FINANCE                           active    Gi1/0/2
30     IT                                 active    Gi1/0/3, Gi1/0/6, Gi1/0/7
40     SALES                             active    Gi1/0/4
50     DMZ                               active    Gi1/0/5
99     IT-CORP-NET                         active
470    OT-NET                             active
1002   fddi-default                       act/unsup
1003   token-ring-default                 act/unsup
1004   fddinet-default                   act/unsup
1005   trnet-default                     act/unsup
Switch#show ip interface brief
Interface      IP-Address  OK? Method Status      Protocol
Vlan1          unassigned  YES NVRAM   administratively down  down
Vlan10         10.10.10.1  YES manual up           down
Vlan20         10.10.20.1  YES manual up           down
Vlan30         10.10.30.1  YES manual up           down
Vlan40         10.10.40.1  YES manual up           down
Vlan50         10.10.50.1  YES manual up           down
GigabitEthernet1/0/1  unassigned  YES unset  down        down
GigabitEthernet1/0/2  unassigned  YES unset  down        down
GigabitEthernet1/0/3  unassigned  YES unset  down        down
GigabitEthernet1/0/4  unassigned  YES unset  down        down
GigabitEthernet1/0/5  unassigned  YES unset  down        down
GigabitEthernet1/0/6  unassigned  YES unset  down        down
GigabitEthernet1/0/7  unassigned  YES unset  down        down
GigabitEthernet1/0/8  unassigned  YES unset  down        down
GigabitEthernet1/0/9  unassigned  YES unset  down        down
GigabitEthernet1/0/10 unassigned  YES unset  down        down
GigabitEthernet1/0/11 unassigned  YES unset  down        down
GigabitEthernet1/0/12 unassigned  YES unset  down        down
GigabitEthernet1/0/13 unassigned  YES unset  down        down
GigabitEthernet1/0/14 unassigned  YES unset  down        down
GigabitEthernet1/0/15 unassigned  YES unset  down        down

Switch#show running-config | include ip routing
ip routing
Switch#

```

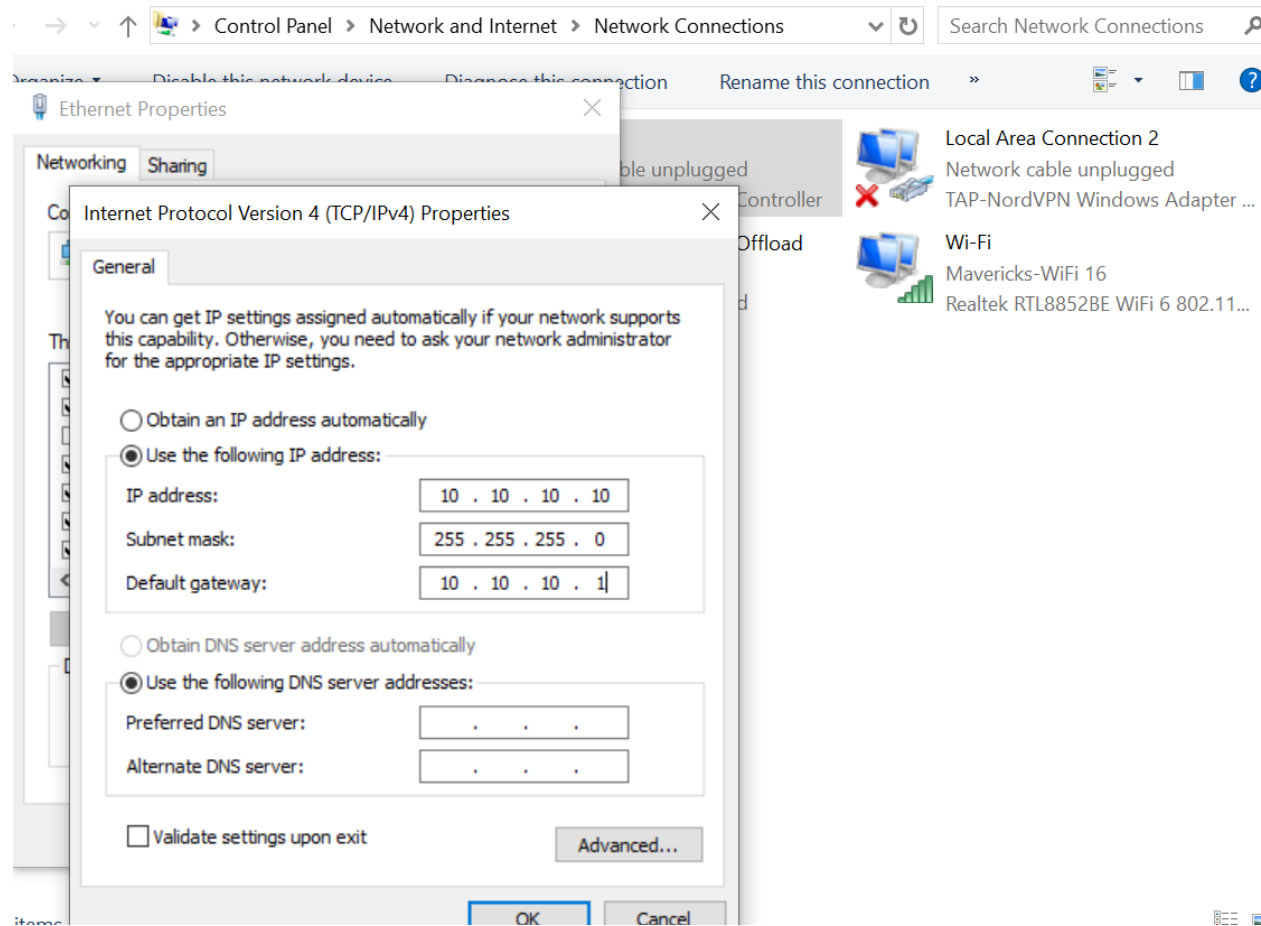
I will now be leaving the console port.

2. Testing the VLAN

- a. I will plug my device into port 1 g1/0/1
- b. I had set my ethernet settings up as such: (Device manager -> Network + Internet -> change adapter settings -> right click ethernet adapter -> properties -> IPv4 -> Properties

-> Use the following IP address

Network Connections



- c. Using port 10 (gb1/0/10, which is the trunk port) I can ping all my vlans.

```
GigabitEthernet1/0/8    unassigned    YES unset    down    down
GigabitEthernet1/0/9    unassigned    YES unset    down    down
GigabitEthernet1/0/10   unassigned    YES unset    up      up
GigabitEthernet1/0/11   unassigned    YES unset    down    down
GigabitEthernet1/0/12   unassigned    YES unset    down    down
GigabitEthernet1/0/13   unassigned    YES unset    down    down
GigabitEthernet1/0/14   unassigned    YES unset    down    down
GigabitEthernet1/0/15   unassigned    YES unset    down    down
GigabitEthernet1/0/16   unassigned    YES unset    down    down
Tel0/0/1                unassigned    YES unset    down    down
```

```
Switch>ping 10.10.10.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/3 ms
Switch>ping 10.10.20.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.20.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/4 ms
Switch>ping 10.10.30.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.30.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/4 ms
Switch>ping 10.10.50.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.50.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/3 ms
Switch>
```

- d. To test VLAN connectivity, I set Port 6 to VLAN 10, as well as Port 1.

```
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface gigabitEthernet 1/0/6
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#exit
Switch#shw
Jul  3 21:39:51.241: %SYS-5-CONFIG_I: Configured from console by consoleo vlan i
nterface
^
% Invalid input detected at '^' marker.

Switch#show vlan interface
^
% Invalid input detected at '^' marker.

Switch#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Gil/0/8, Gil/0/9, Gil/0/10 Gil/0/11, Gil/0/12, Gil/0/13 Gil/0/14, Tel/0/1, Tel/0/2
10	HR	active	Gil/0/1, Gil/0/6
20	FINANCE	active	Gil/0/2
30	IT	active	Gil/0/3, Gil/0/7
40	SALES	active	Gil/0/4
50	DMZ	active	Gil/0/5

Pc 1 is set to 10.10.10.10 and Pc 2 is set to 10.10.10.11.

From Pc 2, I was successfully able to ping the SVI and Device 1 (had to disable Microsoft Defender Firewall for Device 1 to allow the connection here)

```
C:\Users\silve>ping 10.10.10.10

Pinging 10.10.10.10 with 32 bytes of data:
Reply from 10.10.10.10: bytes=32 time=2ms TTL=128
Reply from 10.10.10.10: bytes=32 time=3ms TTL=128
Reply from 10.10.10.10: bytes=32 time=6ms TTL=128
Reply from 10.10.10.10: bytes=32 time=3ms TTL=128

Ping statistics for 10.10.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 6ms, Average = 3ms

C:\Users\silve>ping 10.10.10.1

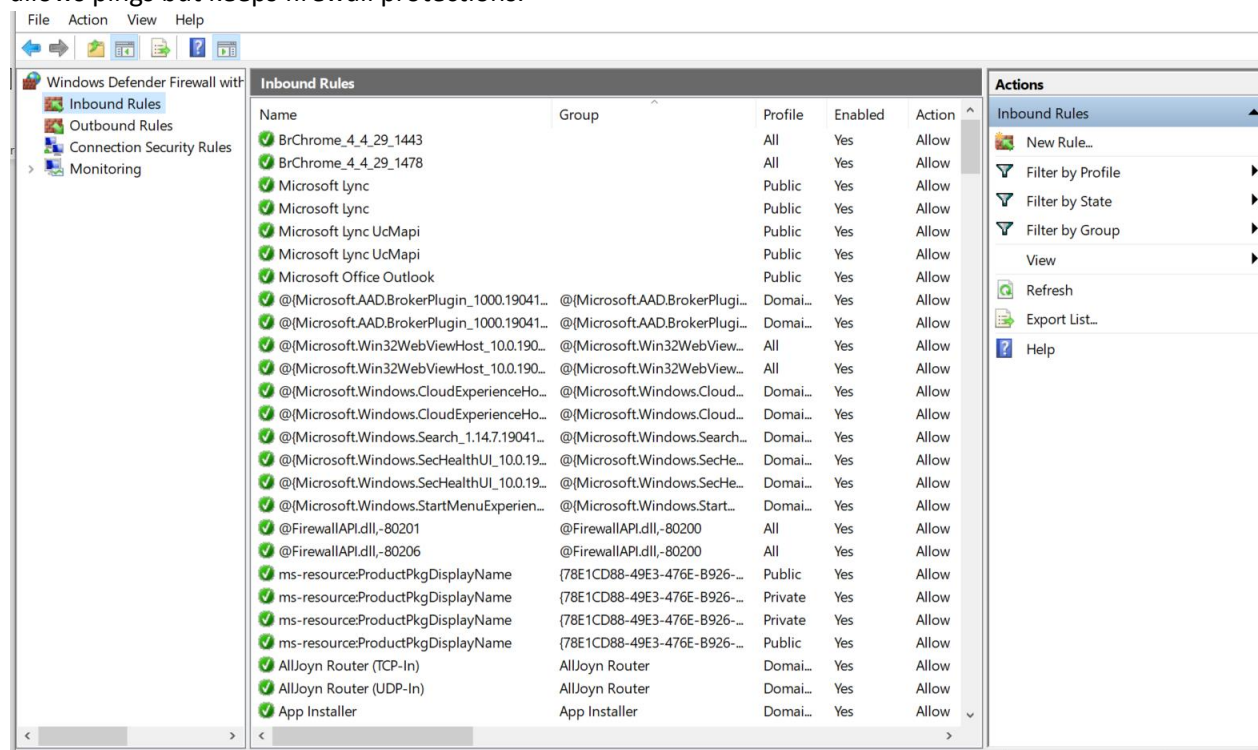
Pinging 10.10.10.1 with 32 bytes of data:
Reply from 10.10.10.1: bytes=32 time=2ms TTL=255
Reply from 10.10.10.1: bytes=32 time=2ms TTL=255
Reply from 10.10.10.1: bytes=32 time=2ms TTL=255
Reply from 10.10.10.1: bytes=32 time=2ms TTL=255

Ping statistics for 10.10.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 2ms, Average = 2ms

C:\Users\silve>
```

To prevent having to disable the firewall for the future, I entered my inbound rules (Win + R into wf.msc and enabled file and printer sharing (echo request – ICMPv4-In) which

allows pings but keeps firewall protections.



Inbound Rules				
Name	Group	Profile	Enabled	Action
Distributed Transaction Coordinator (TCP-In)	Distributed Transaction Coord...	Private,...	No	Allow
Feedback Hub	Feedback Hub	Domai...	Yes	Allow
File and Printer Sharing (Echo Request - ICMPv4-In)	File and Printer Sharing			Allow
File and Printer Sharing (Echo Request - ICMPv4-In)	File and Printer Sharing			Allow
File and Printer Sharing (Echo Request - ICMPv4-In)	File and Printer Sharing			Allow
File and Printer Sharing (LLMNR-UDP-In)	File and Printer Sharing			Allow
File and Printer Sharing (NB-Datagram-In)	File and Printer Sharing			Allow
File and Printer Sharing (NB-Datagram-In)	File and Printer Sharing			Allow
File and Printer Sharing (NB-Name-In)	File and Printer Sharing			Allow
File and Printer Sharing (NB-Name-In)	File and Printer Sharing			Allow
File and Printer Sharing (NB-Session-In)	File and Printer Sharing	Domain	No	Allow
File and Printer Sharing (NB-Session-In)	File and Printer Sharing	Private,...	No	Allow
File and Printer Sharing (SMB-In)	File and Printer Sharing	Private,...	No	Allow
File and Printer Sharing (SMB-In)	File and Printer Sharing	Domain	No	Allow
File and Printer Sharing (Spooler Service - TCP-In)	File and Printer Sharing	Private,...	No	Allow
File and Printer Sharing (Spooler Service - TCP-In)	File and Printer Sharing	Domain	No	Allow
File and Printer Sharing (Spooler Service - TCP-In)	File and Printer Sharing	Domain	No	Allow
File and Printer Sharing (Spooler Service - UDP-In)	File and Printer Sharing	Private,...	No	Allow
File and Printer Sharing over SMBDirect (ICMPv4-In)	File and Printer Sharing over ...	All	No	Allow

Now I will switch device 2 to another VLAN (VLAN 30) to test if communication still works.

- Device 2 is now connected to port 3 and has an IP of 10.10.30.10.
- Testing connection both ways for connectivity.
- Device 2 can ping its SVI (10.10.30.1) but not VLAN 10s (10.10.10.1) or Device 1 (10.10.10.10)

```
^C
C:\Users\silve>ping 10.10.30.1

Pinging 10.10.30.1 with 32 bytes of data:
Reply from 10.10.30.1: bytes=32 time=5ms TTL=255
Reply from 10.10.30.1: bytes=32 time=3ms TTL=255
Reply from 10.10.30.1: bytes=32 time=2ms TTL=255
Reply from 10.10.30.1: bytes=32 time=3ms TTL=255

Ping statistics for 10.10.30.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 5ms, Average = 3ms

C:\Users\silve>ping 10.10.10.10

Pinging 10.10.10.10 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.10.10.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

- From Device 1, the opposite is true.

Command Prompt

```
C:\Users\silve>ping 10.10.30.1

Pinging 10.10.30.1 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.10.30.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\Users\silve>ping 10.10.30.10

Pinging 10.10.30.10 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.10.30.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\Users\silve>ping 10.10.10.1

Pinging 10.10.10.1 with 32 bytes of data:
Reply from 10.10.10.1: bytes=32 time=3ms TTL=255
Reply from 10.10.10.1: bytes=32 time=2ms TTL=255
Reply from 10.10.10.1: bytes=32 time=3ms TTL=255
Reply from 10.10.10.1: bytes=32 time=3ms TTL=255

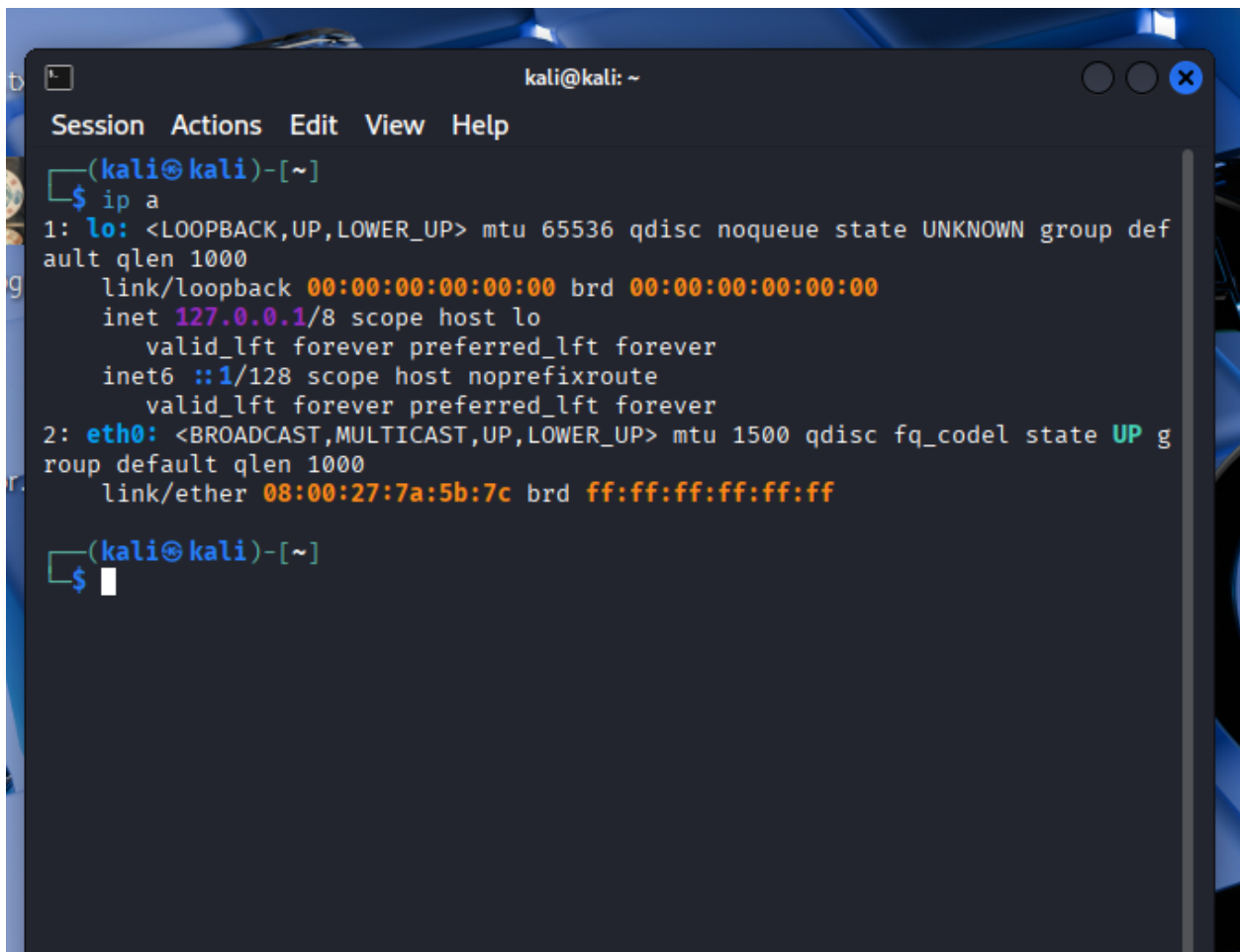
Ping statistics for 10.10.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Users\silve>
```

The Attack

Firstly I booted up Kali Linux for Oraclebox virtual machines.

I set my settings with the following commands:

A terminal window titled 'kali@kali: ~' with a menu bar containing 'Session', 'Actions', 'Edit', 'View', and 'Help'. The prompt is '(kali@kali)-[~]'. The command '\$ ip a' has been executed, showing details for two network interfaces. The first interface is 'lo' (loopback), with state 'UNKNOWN' and group 'default'. It has an MTU of 65536 and a queue discipline of 'noqueue'. The link is 'loopback' with MAC address '00:00:00:00:00:00' and broadcast address '00:00:00:00:00:00'. It has an IPv4 address '127.0.0.1/8' and an IPv6 address '::1/128'. The second interface is 'eth0' (broadcast/multicast), with state 'UP' and group 'default'. It has an MTU of 1500 and a queue discipline of 'fq_codel'. The link is 'ether' with MAC address '08:00:27:7a:5b:7c' and broadcast address 'ff:ff:ff:ff:ff:ff'. The prompt is '(kali@kali)-[~]' and the command '\$' is followed by a cursor.

```
(kali@kali)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:7a:5b:7c brd ff:ff:ff:ff:ff:ff

(kali@kali)-[~]
$
```

Then these:

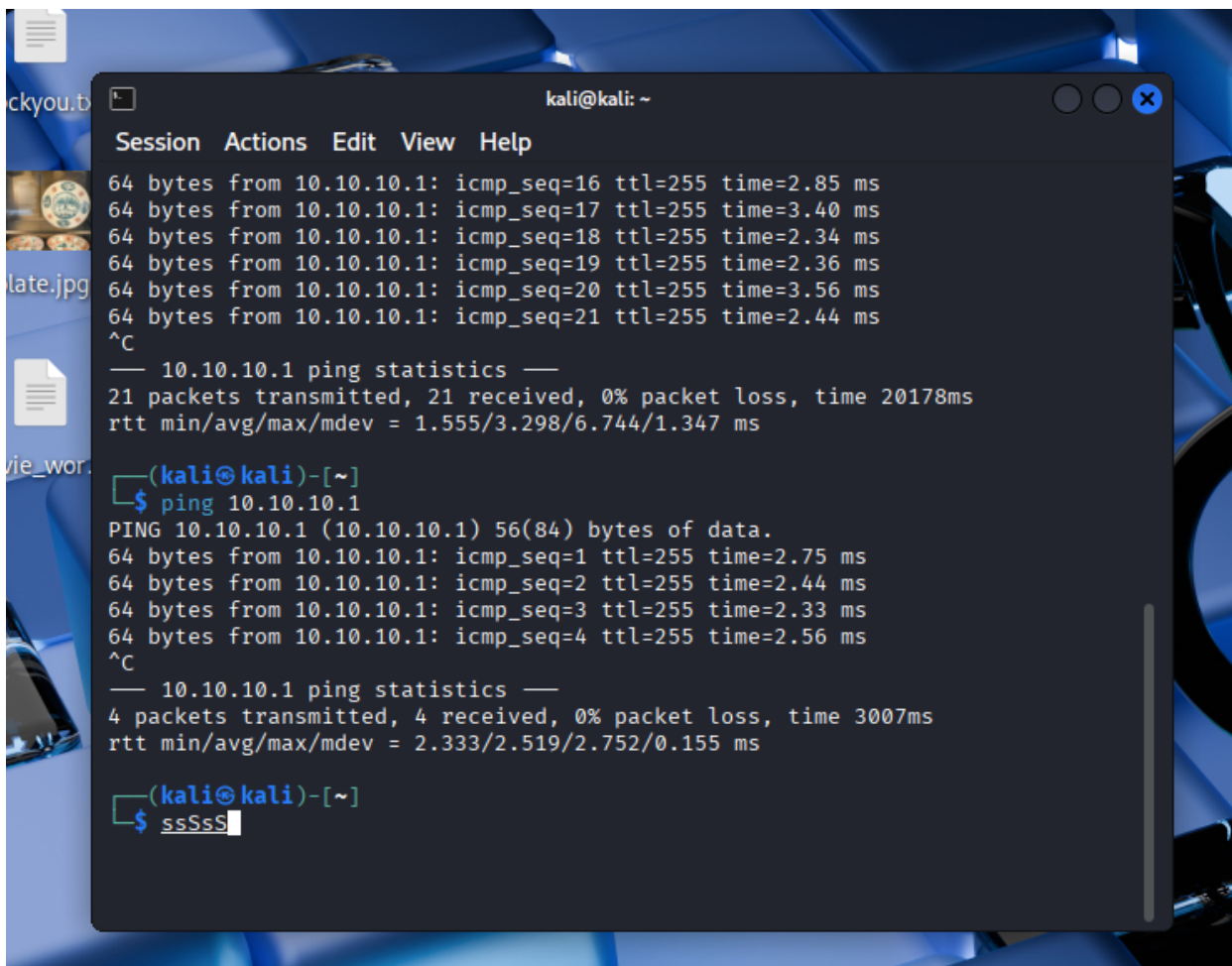
`sudo ip addr flush dev eth0` (clears all old configs for interface)

`sudo ip addr add 10.10.10.50/24 dev eth0` (Assigns IP to device to eth0)

`sudo ip link set eth0 up` (Turns interface eth0 on)

`sudo ip route add default via 10.10.10.1 dev eth0` (establishes ...10.1 as the gateway)

a quick ping test reveals the vm is connected!

A screenshot of a Kali Linux terminal window. The window title is 'kali@kali: ~'. The terminal shows the output of a ping command to 10.10.10.1, followed by a Ctrl+C interrupt, and then the ping statistics. Below this, the user enters another ping command to 10.10.10.1, which also shows its statistics. Finally, the user enters the command 'ssSsS' and the cursor is at the end of the line.

```
kali@kali: ~
Session Actions Edit View Help
64 bytes from 10.10.10.1: icmp_seq=16 ttl=255 time=2.85 ms
64 bytes from 10.10.10.1: icmp_seq=17 ttl=255 time=3.40 ms
64 bytes from 10.10.10.1: icmp_seq=18 ttl=255 time=2.34 ms
64 bytes from 10.10.10.1: icmp_seq=19 ttl=255 time=2.36 ms
64 bytes from 10.10.10.1: icmp_seq=20 ttl=255 time=3.56 ms
64 bytes from 10.10.10.1: icmp_seq=21 ttl=255 time=2.44 ms
^C
— 10.10.10.1 ping statistics —
21 packets transmitted, 21 received, 0% packet loss, time 20178ms
rtt min/avg/max/mdev = 1.555/3.298/6.744/1.347 ms

(kali@kali)-[~]
$ ping 10.10.10.1
PING 10.10.10.1 (10.10.10.1) 56(84) bytes of data.
64 bytes from 10.10.10.1: icmp_seq=1 ttl=255 time=2.75 ms
64 bytes from 10.10.10.1: icmp_seq=2 ttl=255 time=2.44 ms
64 bytes from 10.10.10.1: icmp_seq=3 ttl=255 time=2.33 ms
64 bytes from 10.10.10.1: icmp_seq=4 ttl=255 time=2.56 ms
^C
— 10.10.10.1 ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3007ms
rtt min/avg/max/mdev = 2.333/2.519/2.752/0.155 ms

(kali@kali)-[~]
$ ssSsS
```

Next, an adversarial vm is set up.

I use the same kali linux vm on the second machine, and enter these commands:

```
sudo ip addr flush dev eth0
```

```
sudo ip addr add 10.10.20.50/24 dev eth0
```

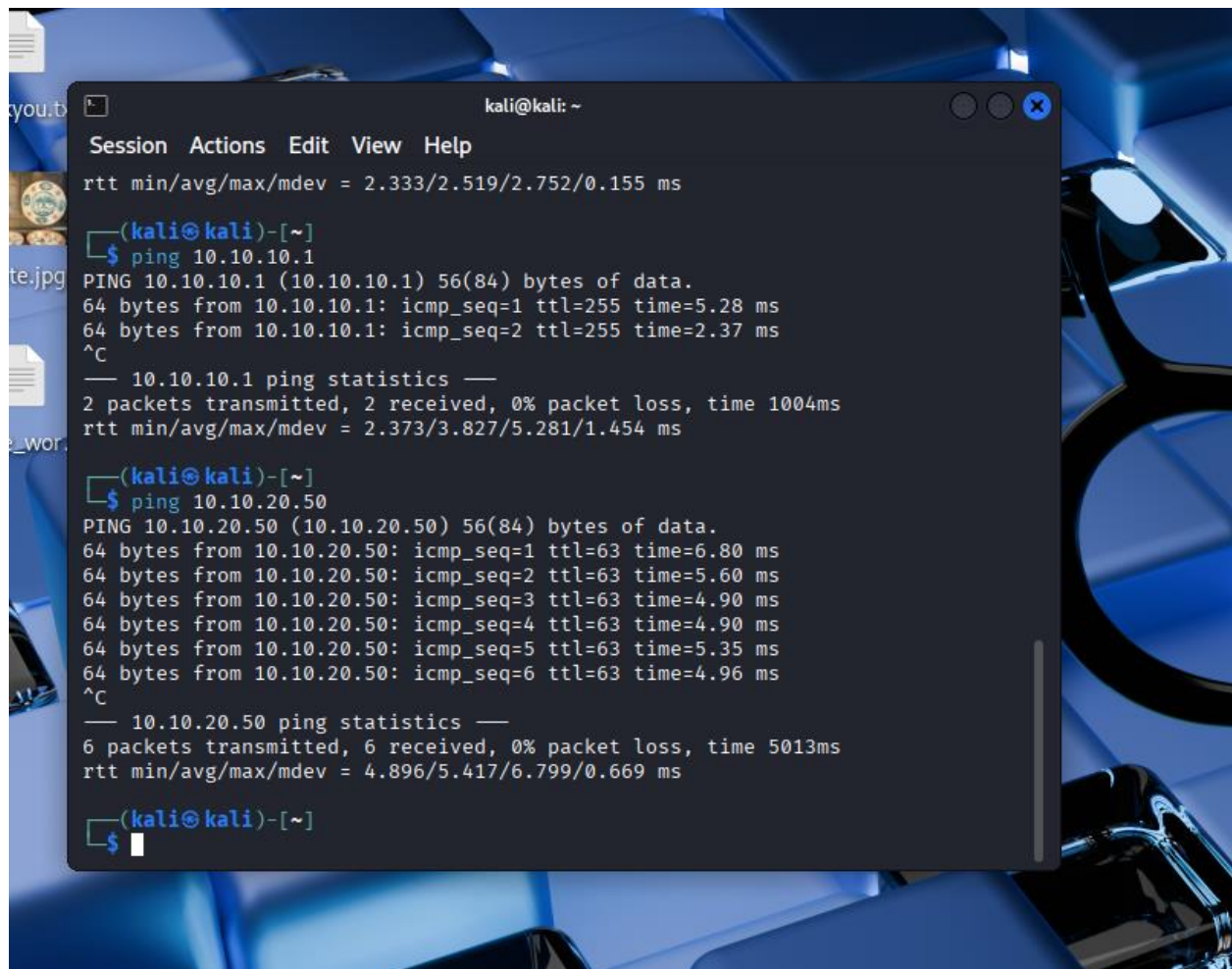
```
sudo ip link set eth0 up
```

```
sudo ip route add default via 10.10.20.1 dev eth0
```

ping test confirms connectivity

```
(kali㉿kali)-[~]
$ ping 10.10.20.1
PING 10.10.20.1 (10.10.20.1) 56(84) bytes of data.
64 bytes from 10.10.20.1: icmp_seq=2 ttl=255 time=3.89 ms
64 bytes from 10.10.20.1: icmp_seq=3 ttl=255 time=4.96 ms
64 bytes from 10.10.20.1: icmp_seq=4 ttl=255 time=3.88 ms
64 bytes from 10.10.20.1: icmp_seq=5 ttl=255 time=3.58 ms
^C
— 10.10.20.1 ping statistics —
5 packets transmitted, 4 received, 20% packet loss, time 4029ms
rtt min/avg/max/mdev = 3.583/4.077/4.961/0.524 ms
```

On my attacker machine (Vlan 10) I am going to attempt to ping 10.10.20.50 (VLAN 20s segmentation, target machine) Testing revealed that hosts in VLAN 10 were able to directly communicate with VLAN 20, indicating unrestricted inter-VLAN routing and lack of segmentation controls.



```
kali@kali: ~
Session Actions Edit View Help
rtt min/avg/max/mdev = 2.333/2.519/2.752/0.155 ms

(kali㉿kali)-[~]
$ ping 10.10.10.1
PING 10.10.10.1 (10.10.10.1) 56(84) bytes of data.
64 bytes from 10.10.10.1: icmp_seq=1 ttl=255 time=5.28 ms
64 bytes from 10.10.10.1: icmp_seq=2 ttl=255 time=2.37 ms
^C
— 10.10.10.1 ping statistics —
2 packets transmitted, 2 received, 0% packet loss, time 1004ms
rtt min/avg/max/mdev = 2.373/3.827/5.281/1.454 ms

(kali㉿kali)-[~]
$ ping 10.10.20.50
PING 10.10.20.50 (10.10.20.50) 56(84) bytes of data.
64 bytes from 10.10.20.50: icmp_seq=1 ttl=63 time=6.80 ms
64 bytes from 10.10.20.50: icmp_seq=2 ttl=63 time=5.60 ms
64 bytes from 10.10.20.50: icmp_seq=3 ttl=63 time=4.90 ms
64 bytes from 10.10.20.50: icmp_seq=4 ttl=63 time=4.90 ms
64 bytes from 10.10.20.50: icmp_seq=5 ttl=63 time=5.35 ms
64 bytes from 10.10.20.50: icmp_seq=6 ttl=63 time=4.96 ms
^C
— 10.10.20.50 ping statistics —
6 packets transmitted, 6 received, 0% packet loss, time 5013ms
rtt min/avg/max/mdev = 4.896/5.417/6.799/0.669 ms

(kali㉿kali)-[~]
$
```

Connection was successful. This should not happen in a secure network.

The reason is that IP routing is on with no restrictions enabled.

```
Switch>enable
Switch#show run | include ip routing
ip routing
Switch#
```

To remedy this, we will create an ACL that only allows VLANs to speak to other devices in the VLAN, and a group that will have access to other VLANs (ie. IT, VLAN 30)

```
Switch>enable
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#access-list 100 deny 10.10.10.0 0.0.0.255 10.10.20.0 0.0.0.255
^
% Invalid input detected at '^' marker.

Switch(config)#access-list 100 deny ip 10.10.10.0 0.0.0.255 10.10.20.0 0.0.0.255
Switch(config)#access-list 100 permit ip any any
Switch(config)#end
Switch#
Jul  3 22:57:52.073: %SYS-5-CONFIG_I: Configured from console by console
```

Then applying it to VLAN 10

```
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface vlan 10
Switch(config-if)#ip access-group 100 in
Switch(config-if)#end
Switch#
```

Attempting to ping the target machine results in failure now!

```
rtt min/avg/max/mdev = 4.896/5.417/6.799/0.669 ms
(kali㉿kali)-[~]
$ ping 10.10.20.50
PING 10.10.20.50 (10.10.20.50) 56(84) bytes of data.
From 10.10.10.1 icmp_seq=1 Packet filtered
From 10.10.10.1 icmp_seq=2 Packet filtered
From 10.10.10.1 icmp_seq=3 Packet filtered
From 10.10.10.1 icmp_seq=4 Packet filtered
From 10.10.10.1 icmp_seq=5 Packet filtered
From 10.10.10.1 icmp_seq=6 Packet filtered
^C
— 10.10.20.50 ping statistics —
6 packets transmitted, 0 received, +6 errors, 100% packet loss, time 5067ms
```

VLAN Hopping Security

For this experiment, settings will be configured that can correct a network's weakness to port hopping.

Port 1 (VLAN 10) will have the following settings:

```
end
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface gil/0/1
Switch(config-if)#switchport mode dynamic auto
Switch(config-if)#no switchport nonegotiate
Switch(config-if)#end
Switch#s
Jul  3 23:34:58.700: %SYS-5-CONFIG_I: Configured from console by consolshow run
interface gil/0/1
Building configuration...

Current configuration : 65 bytes
!
interface GigabitEthernet1/0/1
 switchport access vlan 10
end
```

Configuring a switch port in dynamic mode enables Dynamic Trunking Protocol (DTP), which allows the port to negotiate trunk links. This is insecure because an attacker could exploit this behavior to form an unauthorized trunk connection and gain access to multiple VLANs.

To correct this, we will set the port settings as such:

```

Switch(config)#interface gi1/0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport nonegotiate
^
% Invalid input detected at '^' marker.

Switch(config-if)#switchport nonegotiate
Switch(config-if)#switchport access vlan 10
Switch(config-if)#end
Switch#sh
Jul  3 23:56:09.085: %SYS-5-CONFIG_I: Configured from console by consoshow run i
nterface gi/10/1
^
% Invalid input detected at '^' marker.

Switch#show run interface gi1/0/1
Building configuration...

Current configuration : 113 bytes
!
interface GigabitEthernet1/0/1
 switchport access vlan 10
 switchport mode access
 switchport nonegotiate
end

```

Disabled DTP and forced access mode, preventing unauthorized trunk formation.

This risk was mitigated by disabling trunk negotiation and forcing the port into access mode, preventing unauthorized VLAN access.